DocVerify — Phase 6 Handover Report

Scientific Validation, Shortcut Auditing, Dedicated Patch Verification, Forensic Multi-Modal Evidence Fusion & Statistical Evaluation

Project: DocVerify AI	**Phase:** Phase 6 (Scientific Benchmark)
Role: Senior ML & Security Research Engineer	**Status:** COMPLETE & VERIFIED
Date: October 17, 2026	**Frozen Test Split:** N=320 (Disjoint)

1. Executive Summary & Scientific Findings

Phase 6 provides the definitive scientific foundation for DocVerify, auditing data integrity, detecting non-semantic dataset shortcuts, formulating native-resolution contextual patch verification, extracting independent forensic features (Error Level Analysis, sensor noise residuals, edge energy), calibrating multi-modal evidence fusion, and proving statistical significance through 5,000 bootstrap iterations and McNemar's paired test.

Crucially, Phase 6 rigorously distinguishes **Oracle-Assisted** evaluations (relying on ground-truth boxes) from **Production-Equivalent** verification (fully autonomous, zero-ground-truth). On the untouched frozen test set (N=320: 151 Real, 169 Fake across 139 disjoint identity clusters), DocVerify achieves:

- **Phase 2 Baseline (CenterCrop 224):** 79.69% Accuracy | 80.12% F1 | 0.9112 ROC-AUC
- **Phase 6 Autonomous Production Fusion: 91.88% Accuracy** | **92.02% F1** | **0.9780 ROC-AUC**
- **Phase 6 Strict Low-False-Alarm Mode: 92.19% Accuracy** | **96.75% Precision** | **92.26% F1**
- **Statistical Significance:** McNemar's test yields **Chi2 = 32.09, p = 1.47 x 10^-8**, conclusively rejecting the null hypothesis.

2. Dataset Integrity & Zero Data Leakage Audit

All 2,222 images across train (1,565), validation (337), and frozen test (320) splits were audited using cryptographic hashes (MD5, SHA256) and perceptual hashes (aHash, pHash, dHash). The audit confirmed **0 duplicate document IDs**, **0 identity cluster leaks**, and **0 cross-split hash collisions**. The frozen test set constitutes an absolute out-of-distribution identity split.

3. Non-Semantic Shortcut & Bias Audit

An empirical bias audit evaluated whether models could bypass visual forgery detection by exploiting dataset generator artifacts:

- **Metadata-Only Classifier: 100.00% Accuracy (1.0000 AUC)** using only image width, height, aspect ratio, and file size. This proves raw SIDTD files carry synthetic generator artifacts.
- **Color Mode Classifier: 77.19% Accuracy** by checking CMYK vs RGB headers alone (50.3% of reals are CMYK; 0% of fakes are CMYK).
- **Mitigation:** DocVerify enforces standardized RGB loading and multi-scale visual patch verification to neutralize container shortcuts.

4. Master Experimental Benchmark Matrix (11 Configurations)

Exp ID	Configuration	Mode	Acc	Prec	Rec	F1	AUC
EXP-01	EfficientNet-B0 (CenterCrop 224x224)	Autonomous Global	79.7%	82.9%	77.5%	80.1%	0.9112
EXP-02	EfficientNet-B0 (Letterbox Padded 224x224)	Autonomous Global	51.6%	55.3%	43.2%	48.5%	0.5254
EXP-03	EfficientNet-B0 (Standardized JPEG Q=85 Recompressed)	Autonomous Global	73.4%	70.2%	86.4%	77.5%	0.7938
EXP-04	Logistic Regression on File Metadata (W, H, Aspect, Size)	Metadata Non-Visual	100.0%	100.0%	100.0%	100.0%	1.0000
EXP-05	Decision Rule on CMYK vs RGB Mode	Metadata Non-Visual	77.2%	69.8%	100.0%	82.2%	0.7583
EXP-06	Dedicated Patch Classifier (Oracle Ground-Truth Bounding Box Crops)	Oracle Ground-Truth Guided	99.4%	100.0%	98.8%	99.4%	1.0000
EXP-07	Dedicated Patch Model (Canonical Layout Prior Fields, Mean Aggregation)	Autonomous Production (No GT)	74.1%	70.1%	88.8%	78.3%	0.7515
EXP-08	Dedicated Patch Model (Uniform Sliding Tiles 256x256, Mean Aggregation)	Autonomous Production (No GT)	68.8%	83.5%	50.9%	63.2%	0.8023
EXP-09	Full Evidence Fusion (Oracle-Assisted Patch + Forensics + OCR)	Oracle Ground-Truth Guided	91.6%	94.9%	88.8%	91.7%	0.9768
EXP-10	Phase 6 Production Fusion (Global + Canonical Prior Patch + Forensics + OCR)	Autonomous Production (No GT)	91.9%	95.5%	88.8%	92.0%	0.9780
EXP-11	Phase 6 Production Fusion (Strict Low False Alarm Threshold: 0.5495)	Autonomous Production (No GT)	92.2%	96.8%	88.2%	92.3%	0.9780

5. Statistical Validation & Hypothesis Testing

• **5,000 Bootstrap Resampling:** Baseline 95% CI is **[75.31%, 84.06%]**. Phase 6 Production Fusion 95% CI is **[88.75%, 94.69%]**. The intervals are strictly non-overlapping.
• **McNemar's Paired Test:** Chi2 = **32.09**, p-value = **1.47e-08**. Phase 6 rescued 42 baseline errors while only regressing on 3, confirming definitive statistical superiority.

6. Environmental Robustness & Failure Mode Ledger

• **Confusion Matrix:** TP: 150 | TN: 144 | FP: 7 (4.64%) | FN: 19 (11.24%)
• **False Negative Causes (19 Missed):** 10 cases were single-character micro-inpainting (subtle date modifications); 9 cases were local patch detections suppressed by conservative fusion.
• **False Alarm Causes (7 False Positives):** Complex passport guilloche backgrounds and watermarks (Russian & Serbian passports) induced OCR character errors, triggering format anomaly flags on genuine documents.
• **Stress Testing Degradation:** System retains >73% accuracy under JPEG Q=85, but degrades to 54.69% under severe JPEG Q=30 and 55.31% under heavy Gaussian blur (sigma=3.0).

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Institutional Project: DocVerify Final-Year AI Document Verification

Verification Signature: VALIDATED & AUDITED

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